

dea-tools: Geospatial analysis of satellite data using Xarray, the Open Data Cube, and Digital Earth Australia

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Summary

dea-tools is an open-source Python library that provides a comprehensive set of tools for analysing, visualising, and modelling geospatial and Earth observation data represented as xarray objects. Originally developed to support [Digital Earth Australia \(DEA\)](#) workflows, dea-tools has evolved into a general-purpose library for use with any geospatial dataset. The package includes utilities for spatial and temporal analysis, remote sensing index calculation, machine learning, and interactive visualisation, and integrates seamlessly with the Python geospatial stack (e.g. xarray ([Hoyer & Joseph, 2017](#)), geopandas ([Jordahl et al., 2020](#)), and dask ([Dask Development Team, 2016](#))). Distributed as a pip-installable environment, dea-tools includes JupyterLab and related dependencies, providing a fully managed platform for scientific geospatial analysis across computing environments. Embedded within the broader dea-notebooks ([Krause et al., 2021](#)) repository, the package is supported by extensive documentation and example Jupyter notebooks demonstrating use of dea-tools functionality. Together, dea-tools and dea-notebooks support reproducible and scalable Earth observation science with Python.

Statement of need

Modern Earth observation workflows often involve complex, multi-temporal, and multi-resolution datasets that demand integrated tools for efficient loading, pre-processing, visualisation, and analysis. While foundational libraries such as xarray, geopandas, and dask provide the building blocks, implementing robust analysis workflows and processing pipelines still requires substantial coding effort. The dea-tools library addresses this gap by providing a curated suite of well-documented functions that streamline common geospatial tasks and integrate seamlessly with the Python geospatial stack.

The dea-tools Python package is a comprehensive, open-source toolkit for analysing and interpreting geospatial and Earth observation data, primarily represented as xarray objects. Originally developed by Digital Earth Australia (DEA) to support national-scale satellite data analysis, dea-tools has evolved into a flexible and general-purpose library that supports a wide range of geospatial workflows. It can be deployed across platforms and supports both gridded datasets (e.g. satellite imagery) and vector datasets (e.g. shapefiles or GeoJSON), whether accessed from cloud-based catalogues via Spatio-Temporal Asset Catalog (STAC) or stored locally.

A major advantage of dea-tools is that it comes bundled with all essential dependencies for scientific geospatial analysis. These include [JupyterLab](#), Jupyter notebooks, xarray, geopandas, NumPy ([Harris et al., 2020](#)), dask, and many other libraries from the Python geospatial

ecosystem. When deployed within a virtual environment, dea-tools provides a ready-to-use and pre-configured platform for geospatial analysis, eliminating the need for manual setup or dependency management. This design lowers the barrier to entry for new users by offering a self-contained environment for exploration, visualisation, and large-scale satellite data processing.

dea-tools enables users to:

- Load, combine, and manage satellite or model datasets - including but not limited to DEA products - while handling reprojection and resolution harmonisation;
- Conduct spatial and temporal analyses directly on xarray objects (e.g., extracting temporal statistics, linear regression, and interpolation);
- Apply a large curated list of remote sensing indices such as NDVI, NDWI etc.;
- Perform machine learning and segmentation workflows on geospatial datasets using built-in tools for training, fitting, and prediction that integrates seamlessly with xarray;
- Generate publication-quality visualisations and animations for both static and time-series data;
- Convert between raster and vector formats using utilities that bridge xarray and geopandas;
- Apply domain-specific tools to analyse coastal change, intertidal zones, land cover, wetland and waterbody dynamics, and climate datasets.

The broader dea-notebooks repository, which includes dea-tools, provides extensive Jupyter notebook based documentation. This includes concise how-to guides for specific tasks as well as longer, real-world application examples. These resources feature end-to-end workflows that demonstrate how to acquire satellite data via [odc-stac](#), process it using dea-tools, and visualize outputs interactively within JupyterLab. Together, dea-tools and dea-notebooks offer a reproducible, scalable, and fully managed framework for geospatial analysis in Python, bridging the gap between foundational geospatial libraries and applied EO research.

Features

The following is a non-exhaustive overview of key functionality available in dea-tools. Each function is supported by an accompanying Jupyter notebook that demonstrates its usage.

Data handling

Tools for loading and manipulating both DEA satellite data, along with other providers of geospatial data, are available in the [dea-tools.datahandling](#):

- `load_ard`: Load and combine multiple Geoscience Australia Landsat, Sentinel 2, or Sentinel-1 Analysis Ready Data products using `odc-stac`. Optional features include pixel quality filtering, cloud and contiguity masking, and dropping time steps with a high proportion of poor-quality (e.g. cloudy or shadowed) pixels.
- `load_reproject`: Load and reproject all or part of a raster dataset (stored either locally or remotely) to match the `crs/resolution` of another dataset.

Spatio-temporal analysis tools

Flexible and powerful functionality for scientific analysis with xarray objects:

- Extracting time series statistics, using the module [dea-tools.temporal](#):
 - Generic summary statistics on any time series with `temporal_statistics`;
 - Phenology with `xr_phenology`;
 - Linear regression with `xr_regression`.
- Spatial analysis tools within [dea-tools.spatial](#):
 - Vectorise `xarray.DataArrays` into `geopandas.GeoDataFrames` with `xr_vectorise`;
 - Rasterize `geopandas.GeoDataFrames` into `xarray.DataArrays` with `xr_rasterize`;

- 87 – Extract detailed contours from xarray.DataArrays at subpixel resolution using
- 88 subpixel_contours;
- 89 – Interpolate point data stored in a geopandas.GeoDataFrame into an xarray.DataAr-
- 90 ray with xr_interpolate.

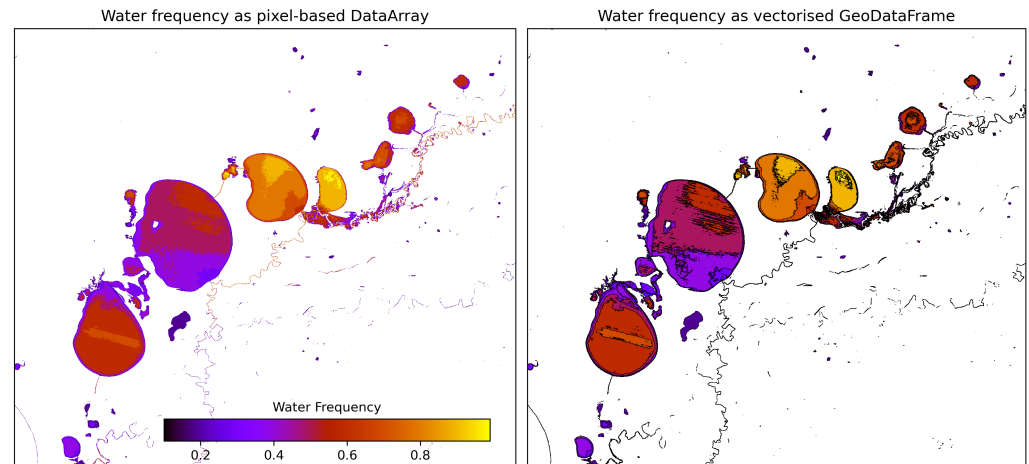


Figure 1: dea-tools enables easy transitions between xarray and geopandas with the `xr_rasterize` and `xr_vectorize` functions. Location is the Menindee Lakes, New South Wales

91 Machine learning tools

92 dea-tools provides a number of functions for simplifying machine learning with xarray:

- 93 ■ Xarray wrappers for the full machine learning pipeline in [dea-tools.classification](#):
 - 94 – prepare data for ML predictions with `sklearn_flatten` and `sklearn_unflatten`;
 - 95 – Fit models with `fit_xr`;
 - 96 – ML predictions with `predict_xr`.
 - 97 – Handle spatial autocorrelation in cross-validation workflows with `spatial_train_test_split`
 - 98 and `SKCV` (spatial k-fold cross validation)
- 99 ■ Validation tools within [dea-tools.validation](#):
 - 100 – Compute common statistical metrics with `eval_metric`;
 - 101 – Generate random points over an xarray object for validation or training using
 - 102 `xr_random_sampling`.

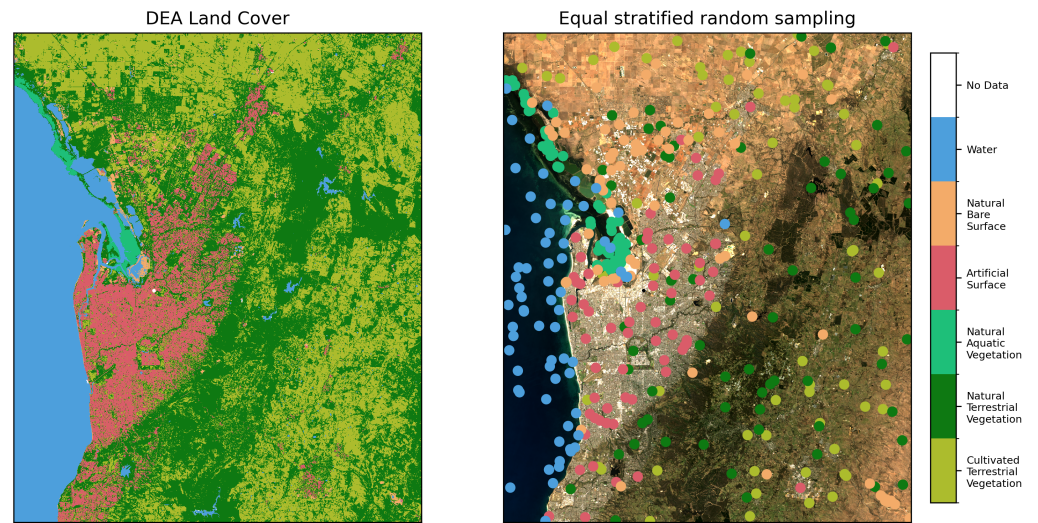


Figure 2: Example extraction of random samples from a classified xarray.DataArray using one of the sampling strategies available in the `xr_random_sampling` function.

Plotting tools

The `dea-tools.plotting` library has extensive functionality for generating beautiful imagery from xarray objects:

- Generate highly customisable animations to visualise changes over time in geospatial datasets using `xr_animation`;
- Create true and false colour composite images from satellite data using `rgb`.



Figure 3: True and false colour composites generated with the `dea-tools.plotting.rgb` function, using DEA annual GeoMAD composite images from 2024. Location is the Norman River, Queensland.

Research projects

`dea-tools` and the broader `dea-notebooks` repository support a wide range of scientific applications. These include but are not limited to agriculture and land cover mapping, wetland, coastal, and surface water monitoring, and climate and fire analysis. The `dea-notebooks` repository has already underpinned more than 25 peer-reviewed studies, demonstrating its robustness and adaptability across domains. Usage of the library is tracked [here](#), providing transparency and insight into its adoption across research projects.

References

- 116
117 Dask Development Team. (2016). *Dask: Library for dynamic task scheduling*. <http://dask.pydata.org>
118
119 Harris, C. R., Millman, K. J., Walt, S. J. van der, Gommers, R., Virtanen, P., Cournapeau, D.,
120 Wieser, E., Taylor, J., Berg, S., Smith, N. J., Kern, R., Picus, M., Hoyer, S., Kerkwijk,
121 M. H. van, Brett, M., Haldane, A., Río, J. F. del, Wiebe, M., Peterson, P., ... Oliphant,
122 T. E. (2020). Array programming with NumPy. *Nature*, 585(7825), 357–362. <https://doi.org/10.1038/s41586-020-2649-2>
123
124 Hoyer, S., & Joseph, H. (2017). xarray: N-D labeled arrays and datasets in Python. *Journal*
125 *of Open Research Software*, 5(1). <https://doi.org/10.5334/jors.148>
126 Jordahl, K., Bossche, J. V. den, Fleischmann, M., Wasserman, J., McBride, J., Gerard,
127 J., Tratner, J., Perry, M., Badaracco, A. G., Farmer, C., Hjelle, G. A., Snow, A. D.,
128 Cochran, M., Gillies, S., Culbertson, L., Bartos, M., Eubank, N., maxalbert, Bilogur,
129 A., ... Leblanc, F. (2020). *Geopandas/geopandas: v0.8.1* (Version v0.8.1). Zenodo.
130 <https://doi.org/10.5281/zenodo.3946761>
131 Krause, C., Dunn, B., Bishop-Taylor, R., Adams, C., Burton, C., Alger, M., Chua, S.,
132 Phillips, C., Newey, V., Kouzoubov, K., Leith, A., Ayers, D., Hicks, A., & DEA Notebooks
133 contributors. (2021). *Digital earth australia notebooks and tools repository*. Geoscience
134 Australia. <https://doi.org/10.26186/145234>